

LAB-1:Use-Case Flow Specification

Problem Statement #63: Community Solar Credit Allocation Manager

NAME:Tejas.V

SRN:-PES1UG25CS850

SEC:I

Use Case ID:-UC-02

Use Case Name:- Transfer Surplus Credits

Primary Actor:- Prosumer Resident

Secondary Actors: Co-op Manager approval flow; Smart Meter Gateway supplies the generation data used to compute balance.

Preconditions:- The prosumer is logged into the portal, has a non-zero surplus credit balance from recorded solar generation, and the target neighbor account exists and is active in the same microgrid co-op.

Postconditions Success:- The prosumer's credit balance is reduced by the transferred amount, the neighbor's monthly billing statement is credited by the same amount, both parties are notified, and the transaction is logged for the monthly billing offset report.

Main Success Scenario:-

1. Prosumer selects “Transfer Credits” from the portal dashboard.
2. System displays the prosumer's current surplus credit balance and a transfer form asking for the neighbor's account ID and the credit amount in kWh.
3. Prosumer enters the neighbor's account ID and the amount of surplus credit to transfer, then submits the request.
4. System validates the requested amount against the prosumer's available balance UC-03: Validate Credit Balance.
5. System deducts the transferred amount from the prosumer's credit balance.
6. System applies the transferred amount as a credit to the neighbor's monthly utility billing statement.

7. System sends a confirmation notification to both the prosumer and the neighbor showing the updated balances.
8. Use case ends successfully.

Alternate Flow — 4a. Insufficient Balance

- 4a1.** System displays an “Insufficient balance” error message and does not alter any balances.
- 4a2.** Prosumer is prompted to enter a lower amount or cancel the request.
- 4a3.** If the prosumer cancels, the use case ends without a transfer; if a new amount is entered, the flow resumes at Main Success Scenario step 4.

Alternate Flow — 4b. Transfer Exceeds Manager-Approval Threshold

- 4b1.** If the validated amount exceeds the configured threshold, the system places the transfer in a “Pending Approval” state instead of proceeding to step 5.
- 4b2.** Co-op Manager reviews the flagged transfer and either approves or rejects it.
- 4b3.** If approved, the flow resumes at Main Success Scenario step 5; if rejected, the prosumer is notified and the use case ends without a transfer.